Agent

课程项目说明文档

1. 项目目标

Agent

2. 课程要求对应

3. 使用场景

“ ”“ ” Agent
Agent

Demo CRM VIN API

4. V1–V4 迭代

5. 技术架构

Python `search_records`
`analyze_trend` `retrieve_evidence` `link_entities` `create_followup` RAG ReAct/
Agent Markdown JSON JSONL

Agent → → → → → → 8 2

`utility = expected_information_gain × source_reliability − retrieval_cost − duplication_penalty`

6. 数据与案例

Honda/Acura EA23003 2023-11-28
URL SHA-256

NHTSA FLAT_INV NHTSA API 2023 Honda Civic CIVIC SEDAN CIVIC HATCHBACK CIVIC SEDAN SI CIVIC HATCHBACK TYPE R Complaints API HTTP 400 count=0 `EMPTY_RESPONSE_UNVERIFIED`

7. 实验与结果

RAG ReAct/
Agent

88 Qwen API 3.125/5 RAG 2.125/5 ReAct/ 3.250/5 Agent 3.193/5 Agent RAG ReAct V4 24 Agent 66.7% 16.7% RAG 50% 16.7% 100% 264

token
Agent

8. Demo 运行与演示脚本

python -m venv .venv

source .venv/bin/activate

pip install -r requirements.txt

python webapp.py

`http://127.0.0.1:8765` EA23003 “ ”

DeepSeek API Key `.env.local` `MODEL_API_KEY` `MODEL_BASE_URL=https://api.deepseek.com` `MODEL_NAME=deepseek-chat` API Agent

9. 能力边界与企业接入

Agent

Demo API

10. 交付清单

- GitHub `v1-baseline` `v2-data-contracts` `v3-dynamic-policy` `v4-final-demo` `main`
- `README.md` API Demo
- `outputs/vehicle-defect-agent-final.pptx` 12
- `docs/project-report.md` Word/PDF
- `enterprise_demo.html` `webapp.py` Demo
- `data/public_demo/`
- `data/eval/` `docs/*results*.md`
- `tests/` NHTSA

11. AI 协作与纠错记录

1. API HTTP 400 API
2. Agent
3. Agent V4 pilot
4. Demo

12. 参考资料

- Yao et al., ReAct, 2022: https://arxiv.org/abs/2210.03629
- Schick et al., Toolformer, 2023: https://arxiv.org/abs/2302.04761
- Asai et al., Self-RAG, 2023: https://arxiv.org/abs/2310.11511
- Yan et al., CRAG, 2024: https://arxiv.org/abs/2401.15884
- Shinn et al., Reflexion, 2023: https://arxiv.org/abs/2303.11366
- NHTSA datasets and APIs: https://www.nhtsa.gov/nhtsa-datasets-and-apis

版本路线

V1			
V2			
V3			pilot
V4			88

主实验结果

	3.125 / 5
RAG	2.125 / 5

ReAct/	3.250 / 5
Agent	3.193 / 5